path meaningless.”

This is not terribly different from the double-slit experiment—indeed, this is the double-slit experiment, just with a slightly different geometry. And as with the double-slit experiment, it’s tempting to say that the photon can tell whether the second beam splitter is there before it starts on its journey through the experiment. If there’s only one beam splitter, the photon travels down only one path. But if the second beam splitter is in place, the photon travels down both paths, so it can interfere with itself.

But Wheeler introduced one more twist into this experiment: the delayed choice. There’s some distance between the beam splitter and the mirror on the lower right-hand side (Figure A.1a). Let’s make that distance big—several kilometers, say. In that case, it’ll take the photon, traveling at the speed of light, about a dozen microseconds to get from the beam splitter to the detectors. That gives us plenty of time to direct a computer to insert (or remove) the second beam splitter *after* the photon leaves the first beam splitter. In other words, we can delay the choice of which experiment we want to do—Figure A.1a or Figure A.1b—until after the photon is already on its way through the experiment. Yet, if we do this, our results are no different. When the second beam splitter is in place, the photon will never arrive at detector 2. And when the second beam splitter is removed, the photon will arrive at each detector about half of the time.

These results are quite strange—yet they have been confirmed by actual experiments. Clearly, this happens. But how can the photon “decide” whether to travel down just one path after it’s already passed through the first beam splitter? We can make the apparent paradox even worse by increasing the distance the photon travels. In principle, there’s no reason this experiment can’t be done using a setup a light-year long, or even billions of light-years. It’s as if a photon can edit its own past in addition to sometimes being in two places at once—or as if our own choices about our experimental setup can change the distant past. And indeed, Wheeler endorsed this view, saying that “we must conclude that our very act of measurement not only revealed the nature of the photon’s history on its way to us, but in some sense *determined* that history. The past history of the universe has no more validity than is assigned by the measurements we make—now!”

But this is just one view of this experiment, using Wheeler’s version of